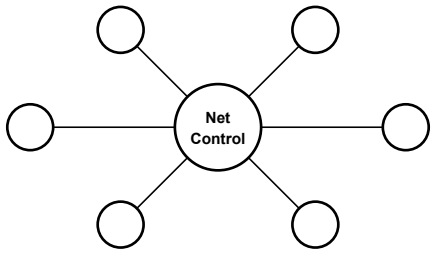
Tech Dessert #6: Public Service and Emergency Operating

Companion reference for meeting 6 -- nets, traffic handling, and the emergency-service organizations ARES and RACES.

Group T2C · 12 pool questions · Self-study handout · Practice exams at eham.net/exams

This Dessert covers one of the best reasons to be a ham: helping your community when normal communications fail. It is a reference sheet – read it, study the net diagram, then drill the question bank.



Net Control directs the net; you transmit only when called.

A net is an organized on-air meeting. One station – Net Control – runs it, and the others transmit only when called on.

Public Service and Emergency Operating

Topic	What to know	Q
Do FCC rules ever stop applying	No – **FCC Part 97 rules always apply**, in every situation.	T2C01
What ARES is	The **Amateur Radio Emergency Service**: licensed amateurs who volunteer their skills and equipment for public service.	T2C06
What RACES is	An FCC Part 97 service for **civil-defense communications** during national emergencies.	T2C04
What RACES requires	**Certification by a civil defense agency.**	T2C12
Duties of a Net Control Station	**Calls the net to order** and **directs communications** between stations checking in.	T2C02
Standard practice when you are in a net	**Transmit only when directed** by Net Control – unless you are reporting an emergency.	T2C07
What “traffic” means in net operation	**Formal messages** exchanged by net stations.	T2C05
The “preamble” of a formal message	The information needed to **track the message**.	T2C10
“Check” in a radiogram header	The **number of words** in the message’s text.	T2C11
Getting unusual words across correctly	**Spell them** with a standard phonetic alphabet.	T2C03
What Winlink does	Relays messages using **email addresses based on amateur call signs**.	T2C08
Operating outside your license privileges	Permitted **only** for the immediate **safety of life or protection of property**.	T2C09

Remember: in an emergency, amateur radio shines – but **FCC rules still apply**. The two emergency organizations are **ARES** (volunteer amateurs, registered for public service) and **RACES** (civil-defense communications, needs civil-defense certification). On a **net**, one station – **Net Control** – runs the show, and you transmit only when called. You may step outside your license privileges **only** to protect human life or property.

Dessert 6 Question Bank (T2C)

T2C01. When do the FCC Part 97 Amateur Radio Service rules NOT apply to the operation of an amateur station?
A. When operating under RACES rules
B. When operating under FEMA rules
C. When operating under ARES rules
D. FCC rules always apply

T2C02. Which of the following are typical duties of a Net Control Station?
A. Choose the regular net meeting time and frequency
B. Ensure that all stations checking into the net are properly licensed for operation on the net frequency
C. Call the net to order and direct communications between stations checking in
D. All these choices are correct

T2C03. What technique is used to ensure that voice messages containing unusual words are received correctly?
A. Send the words by voice and Morse code
B. Use the Q-code “QSR” to ask the receiving station to repeat the words back
C. Spell the words using a standard phonetic alphabet
D. All these choices are correct

T2C04. What is RACES?
A. An emergency organization combining amateur radio and citizens band operators and frequencies
B. An international radio experimentation society
C. A radio contest held in a short period, sometimes called a “sprint”
D. An FCC Part 97 amateur radio service for civil defense communications during national emergencies

T2C05. What does the term “traffic” refer to in net operation?
A. Formal messages exchanged by net stations
B. The number of stations checking in and out of a net
C. Operation by mobile or portable stations
D. A count of the number of activations of the net each month

T2C06. What is the Amateur Radio Emergency Service (ARES)?
A. A group of licensed amateurs who have voluntarily registered their qualifications and equipment for communications duty in the public service
B. A group of licensed amateurs who are members of the military and who voluntarily agreed to provide message handling services in the case of an emergency
C. A training program that provides licensing courses for those interested in obtaining an amateur license to use during emergencies
D. A training program that certifies amateur operators for membership in the Radio Amateur Civil Emergency Service

T2C07. Which of the following is standard practice when you participate in a net?
A. When first responding to the net control station, transmit your call sign, name, and address as in the FCC database
B. Record the time of each of your transmissions
C. Unless you are reporting an emergency, transmit only when directed by the net control station
D. All these choices are correct

T2C08. Which of the following relays messages using email addresses based on amateur callsigns?
A. Winlink
B. FT8
C. PSK31
D. AMTOR

T2C09. Are amateur station control operators ever permitted to operate outside the frequency privileges of their license class?
A. No
B. Yes, but only when part of a FEMA emergency plan
C. Yes, but only when part of a RACES emergency plan
D. Yes, but only in situations involving the immediate safety of human life or protection of property

T2C10. What information is contained in the preamble of a formal traffic message?
A. The email address of the originating station
B. The address of the intended recipient
C. The telephone number of the addressee
D. Information needed to track the message

T2C11. What is meant by “check” in a radiogram header?
A. The number of words or word equivalents in the text portion of the message
B. The call sign of the originating station
C. A list of stations that have relayed the message
D. A box on the message form that indicates that the message was received and/or relayed

T2C12. Which of the following requires certification by a civil defense agency?
A. ARES
B. RACES
C. MARS
D. SKYWARN

Answer Key

T2C01: D	T2C02: C	T2C03: C	T2C04: D
T2C05: A	T2C06: A	T2C07: C	T2C08: A
T2C09: D	T2C10: D	T2C11: A	T2C12: B